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Faculty of Informatics and Information Technology
Study programme: Information Security

Remote fault attacks using DVFS technique
Research Proposal for a Master's Thesis

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1 Introduction

A fault injection attack is an attack on an electronic device in order to induce faults leading to security failures, leaking critical information. These attacks leverage lasers, electromagnetism or other means to stress the system. The faults consist of operation skips, operation repetitions or bit flips. A successful attack can have destructive consequences, such as bypassing password verification, breaking digital signatures or hash functions, or Denial of Service (DoS). These attacks, while effective, require access to the victim machine and (in the case of laser fault attacks) can be expensive [1].

Remote fault attacks aim to improve on this by not requiring direct access to the machine, leveraging remote means to induce faults in the system instead. Unlike traditional fault injection attacks, remote fault attacks are more practical and cost-effective. The consequences of a remote fault attack can be the same as of their traditional counterparts.

Dynamic Voltage and Frequency Scaling (DVFS) is a technique that dynamically changes the processors voltage and frequency. Utilized extensively in mobile devices and battery operated devices to extend battery life, but also in quiet computing systems and cloud computation, to reduce cooling costs. There is no doubt, that this technology became an irreplaceable part of processor architecture.

2 Motivation

DVFS introduces a critical security problem. Unrestrained software access to this energy management hardware is possible from lower privileged software [2]. This can be exploited to induce faults in a processor remotely, transforming a traditional fault attack into a cheap, practical and highly effective threat.

It is the job of security researchers to find vulnerabilities and attack vectors in new technology, so that when found, they are made public and safeguards can quickly be put in place. It is always better for a vulnerability to be found by a friendly entity, rather than a hostile attacker. By developing a better and more reliable exploit, we further speed up the development of new security features.

This work seeks to address this problem by investigating these vulnerabilities and demonstrating them on two real world applications. Given the critical role DVFS plays in modern computing systems, the limited number of studies on this attack leaves a significant hole in security knowledge. While there is no public knowledge about this attack being exploited with malicious intent, it is only a matter of time before someone attacks an unprepared service or institution.

3 Problem analysis

DVFS, while a great addition, unintentionally provides attackers with a new attack vector. By exploiting the software controls of DVFS, an attacker can push the voltage and frequency past the vendor recommended limits. When the processor operates under such extreme circumstances, hardware faults occur.

Inducing these hardware faults is a tricky subject on its own, but inducing them effectively is a whole another story. The attackers usually need to isolate the fault to a single processor core and to avoid crashing the system, a precise timing is necessary. Existing implementations of this attack managed to induce an accurate fault during an AES cryptography calculation to infer AES keys based on pairs of correct and faulty plaintext [2]. By causing a single-byte random corruption during the seventh round of AES decryption, they managed to reduce the number of AES-128 key hypotheses from 2^{128} down to 2^{12} , enabling a brute-force search for the key. The same researchers also managed to utilize this attack (called CLKscrew) to load self-signed applications into ARM TrustZone.

Another successful implementation of a fault attack (called **DeepLaser**) was done by a group of researchers and targeted four widely used activation functions in Deep Neural Networks (DNNs) development - ReLu, softmax, sigmoid, and tanh [3]. The researchers used a laser fault injection and noted, that this approach might not be possible in many scenarios and suggested simpler ways to disturb the circuits, such as voltage/clock glitching.

Another paper writes about a group of researchers that managed to circumvent Intel software guard extensions (SGX). SGX is a set of instruction codes present on some Intel CPUs. These instructions allow user-level and os-level code to define protected areas of memory called enclaves. Data and code originating from an enclave is stored encrypted and are decrypted by the processor at runtime. This ensures protection from being read by other code, including code being ran at kernel-level. The researchers successfully implemented a DVFS remote fault attack (called VoltJockey) and managed to extract the key of an AES executed in an enclave [4].

The same researchers published another paper, where they use the same attack (VoltJockey) to break ARM TrustZone. This attack focuses on over-volting the CPU, rather than overclocking, as was the case in the CLKscrew attack. This makes the attack harder to detect, as hardware frequency locking works for overclocking only [5]. The result of this attack, was once again an AES key extraction. They also managed to breach the RSA-based TrustZone authentication, enabling the loading of untrusted applications into TrustZone.

4 Research objectives

My objectives in this work will consist of implementing a remote fault attack utilizing the DVFS technology. I will specifically target a neural network with the aim to induce misclassification. I will also target the ascon lightweight cryptography cipher recently selected by US National Institute of Standards and Technology for standardization [6].

- RQ1: How can a remote fault attack induce misclassification in DNNs and to what extent?
- RQ2: How resilient is the ascon lightweight cryptography cipher to remote fault attacks and how can this attack compromise such a cipher?
- RQ3: What are the implications of remote fault attacks using DVFS and how can they be mitigated?

5 Work plan

- 2. (summer) semester - DPI
 - Further analysis of the problem at hand
- 3. (winter) semester - DPII
 - Dedicated to experimenting and implementing the attack
- 4. (summer) semester - DPIII
 - Finalizing experiments
 - Conducting tests
 - Finalizing the document and the work as a whole

6 Master’s Thesis assignment

Remote fault attacks utilizing Dynamic Voltage and Frequency Scaling (DVFS) technology present a new attack vector in modern computing systems. Analyze current solutions to these attacks and assess their efficiency and potential impact on two applications: neural networks and a cryptographic cipher Ascon. Investigate and describe specific vulnerabilities in neural networks from the perspective of inducing misclassifications and in Ascon from the perspective of compromising security. Demonstrate a remote fault attack on both targets and evaluate their efficiency and reliability. Compare the results with existing implementations and assess their implications for the security of modern systems. Based on the research findings, propose countermeasures and safeguards, that can enhance the resilience of modern systems against these attacks.

Literature:

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